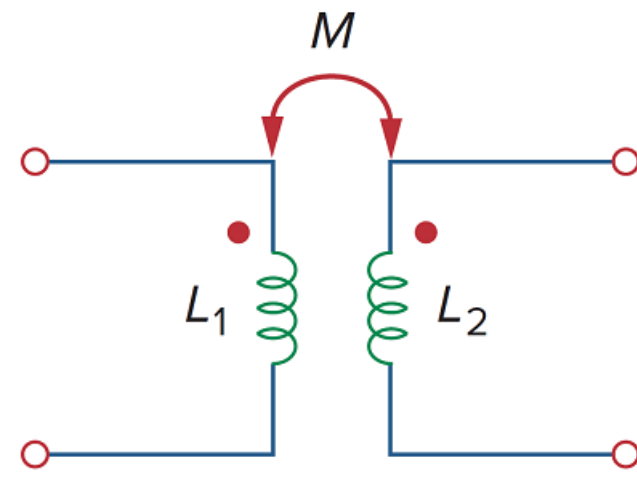


Problem

Using Fig. 13.100, design a problem to help other students better understand linear transformers and how to find T-equivalent and Π -equivalent circuits.

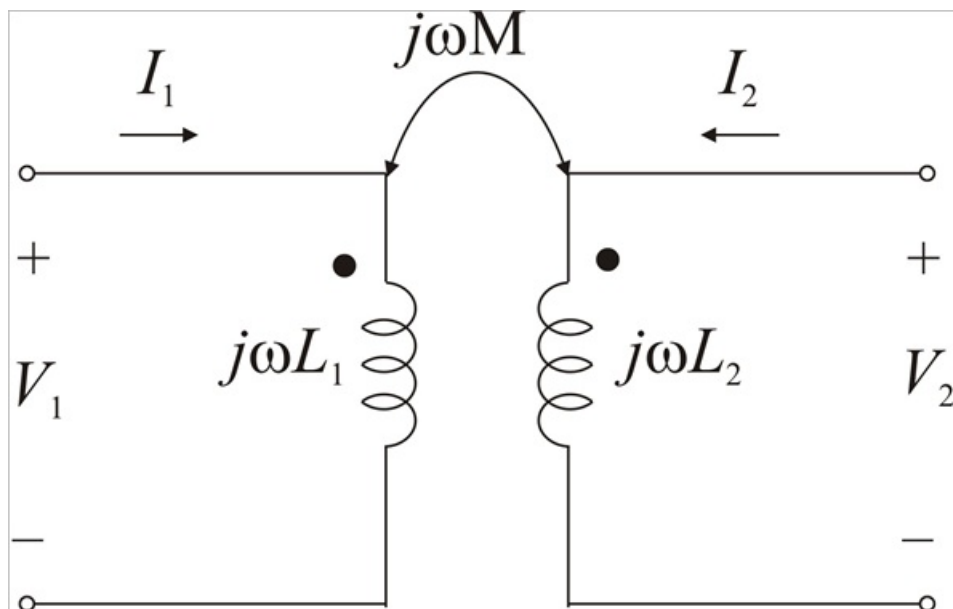
Figure 13.100



Step-by-step solution

Step 1 of 6

The linear transformer equivalent circuit is



Step 2 of 6

Applying KVL to mesh 1,

$$-V_1 + j\omega L_1 I_1 + j\omega M I_2 = 0$$

$$V_1 = j\omega L_1 I_1 + j\omega M I_2 \dots\dots\dots (1)$$

Applying KVL to mesh 2,

$$-V_2 + j\omega L_2 I_2 + j\omega M I_1 = 0$$

$$V_2 = j\omega M I_1 + j\omega L_2 I_2 \dots\dots\dots (2)$$

Equation (1) and (2) expressed in matrix form as

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots\dots (3)$$

By using matrix inversion method,

$$I = X^{-1}V$$

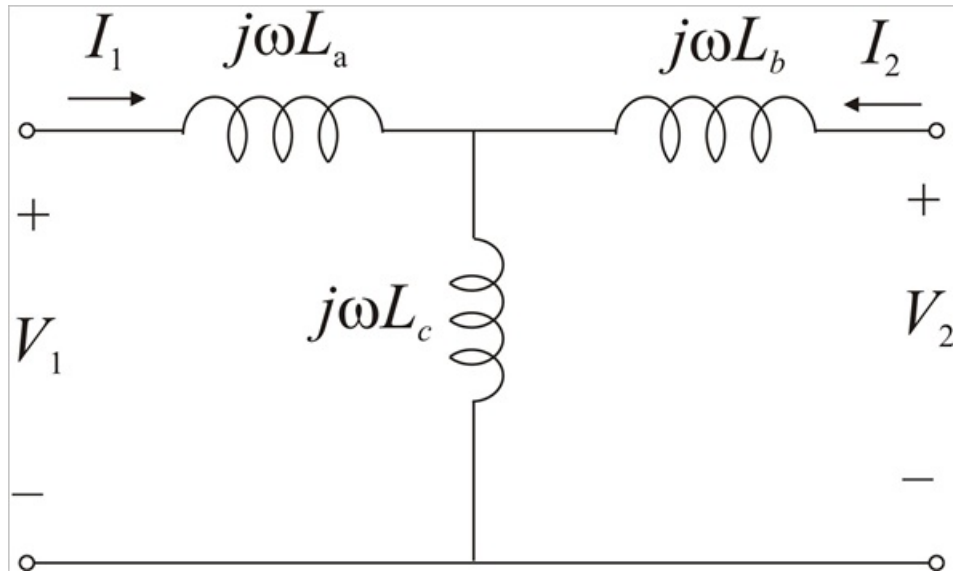
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{(j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)} \begin{bmatrix} j\omega L_2 & -j\omega M \\ -j\omega M & j\omega L_1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \begin{bmatrix} \frac{j\omega L_2}{(j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)} & \frac{-j\omega M}{(j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)} \\ \frac{-j\omega M}{(j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)} & \frac{j\omega L_1}{(j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)} \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{L_2}{j\omega(L_1 L_2 - M^2)} & \frac{-M}{j\omega(L_1 L_2 - M^2)} \\ \frac{-M}{j\omega(L_1 L_2 - M^2)} & \frac{L_1}{j\omega(L_1 L_2 - M^2)} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \dots\dots\dots (4)$$

Step 3 of 6

Case (1) T -equivalent circuit is shown below



Step 4 of 6

Applying KVL to mesh 1,

$$-V_1 + j\omega L_a I_1 + j\omega L_c (I_1 + I_2) = 0$$

$$V_1 = j\omega (L_a + L_c) I_1 + j\omega L_c I_2 \dots\dots\dots (5)$$

Applying KVL to mesh 2,

$$-V_2 + j\omega L_b I_2 + j\omega L_c (I_2 + I_1) = 0$$

$$V_2 = j\omega L_c I_1 + j\omega (L_b + L_c) I_2 \dots\dots\dots (6)$$

Equations (5) and (6) expressed in matrix form as,

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega (L_a + L_c) & j\omega L_c \\ j\omega L_c & j\omega (L_b + L_c) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots\dots (7)$$

Comparing equations (3) and (7)

$$j\omega L_1 = j\omega (L_a + L_c)$$

$$L_1 = L_a + L_c$$

$$L_a = L_1 - L_c \dots\dots\dots (8)$$

And

$$j\omega L_2 = j\omega (L_b + L_c)$$

$$L_2 = L_b + L_c$$

$$L_b = L_2 - L_c \dots\dots\dots (9)$$

$$j\omega M = j\omega L_c$$

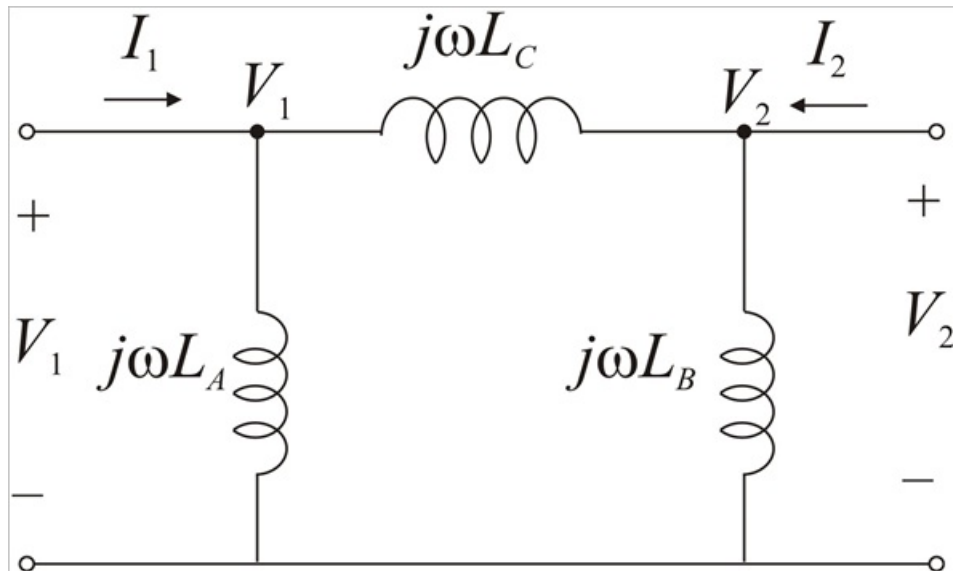
$$\therefore \boxed{L_c = M}$$

Substituting value of L_c in equations (8) and (9), we get

$$\therefore \boxed{L_a = L_1 - M}$$

$$\therefore \boxed{L_b = L_2 - M}$$

Case (2) Π – Equivalent circuit is shown below



Applying KCL at node V_1 ,

$$I_1 = \frac{V_1}{j\omega L_A} + \frac{V_1 - V_2}{j\omega L_C}$$

$$I_1 = V_1 \left[\frac{1}{j\omega L_A} + \frac{1}{j\omega L_C} \right] - V_2 \left[\frac{1}{j\omega L_C} \right] \dots\dots\dots (10)$$

Applying KCL at node V_2

$$I_2 = \frac{V_2}{j\omega L_B} + \frac{V_2 - V_1}{j\omega L_C}$$

$$I_2 = -V_1 \left[\frac{1}{j\omega L_C} \right] + V_2 \left[\frac{1}{j\omega L_B} + \frac{1}{j\omega L_C} \right] \dots\dots\dots (11)$$

Equation (10) and (11) expressed in matrix form as,

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{j\omega L_A} + \frac{1}{j\omega L_C} & -\frac{1}{j\omega L_C} \\ -\frac{1}{j\omega L_C} & \frac{1}{j\omega L_B} + \frac{1}{j\omega L_C} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \dots\dots\dots (12)$$

Step 6 of 6

Comparing equations (4) and (12), we get

$$\frac{L_2}{j\omega(L_1 L_2 - M^2)} = \frac{1}{j\omega L_A} + \frac{1}{j\omega L_C}$$

$$\frac{L_2}{(L_1 L_2 - M^2)} = \frac{L_C + L_A}{L_A L_C} \dots\dots\dots (13)$$

$$\frac{L_1}{j\omega(L_1 L_2 - M^2)} = \frac{1}{j\omega L_B} + \frac{1}{j\omega L_C}$$

$$\frac{L_1}{(L_1 L_2 - M^2)} = \frac{L_C + L_B}{L_C L_B} \dots\dots\dots (14)$$

$$\frac{-M}{j\omega(L_1 L_2 - M^2)} = \frac{-1}{j\omega L_C}$$

$$\therefore L_C = \frac{L_1 L_2 - M^2}{M}$$

Substituting value of L_C in equations (13) and (14), we get

$$\left(\frac{L_1 L_2 - M^2}{M} \right) L_A = \left(\frac{L_1 L_2 - M^2}{M} + L_A \right) \left(\frac{L_1 L_2 - M^2}{L_2} \right)$$

$$L_A \left(\frac{1}{M} - \frac{1}{L_2} \right) = \frac{L_1 L_2 - M^2}{L_2 M}$$

$$\therefore L_A = \frac{L_1 L_2 - M^2}{L_2 - M}$$

Similarly,

$$\therefore L_B = \frac{L_1 L_2 - M^2}{L_1 - M}$$